

The Milky Way on a phone

Print it, fold it, take it out with you.

Fragments of the Universe
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THE TWO THINGS THAT DECIDE THE NIGHT

A tripod

Night photography holds the shutter open for several seconds, and anything hand-held moves in that time. A small tabletop tripod is enough. Bonus: many phones only unlock their longest exposures once they are sure the phone is completely still.

A dark sky

From a city the skyglow is brighter than the Milky Way, and no amount of editing recovers what the phone never saw. An hour out of most cities is usually enough, and the difference is dramatic rather than subtle.

YOUR PHONE, BY MAKE

iPhone

Lens

Main wide lens, not the ultra-wide or the zoom. Biggest sensor, fastest aperture.

Up to iPhone 16

The moon icon appears in the dark. On a tripod the exposure slider runs to about 30 seconds. Drag it to maximum and tap the shutter.

iPhone 17 series

The slider is gone. Night mode is Off, Auto or Max. Choose Max: up to 30s on a tripod. Auto will not give you a long exposure.

ProRAW

Turn it on if your model offers it. It records a file you can brighten properly afterwards.

Samsung Galaxy

Pro mode

Manual shutter to 30 seconds, plus manual ISO and focus. Enough on its own.

The hidden astro mode

Download Expert RAW from the Galaxy Store. Open it, tap the settings icon, turn on Special photo options. An astrophotography icon then appears.

If you cannot find it

It only shows at 12MP resolution. Listed for S22 and S23 series onwards.

Star trails

Hyperlapse mode, speed 300x, then a star trails toggle appears. Set resolution to UHD.

Pixel and other Android

Pixel

Night Sight, on a tripod, then leave it alone. It relabels itself as astrophotography and offers up to about four minutes, stacking frames internally.

Everything else

Most Android phones have a Pro mode somewhere in the camera app. 30 seconds is plenty.

Worth ten minutes

Go through every mode in your camera app before deciding your phone cannot do this. The useful one is rarely on the front screen.

IF YOUR PHONE HAS A PRO MODE, SET THIS

ISO

3200

Until the sky looks grey rather than black, then back off a step

SHUTTER

20"

Long enough to record the core, short enough to repeat all night

WB

4000K

Consistent colour across every frame of the night

FOCUS

Infinity

The step most people skip. Slide it all the way by hand

FORMAT

RAW

Real latitude to lift the shadows afterwards

TIMER

3s

Stops the tap itself shaking the phone

No manual mode? You lose very little. Night mode on a tripod, fired again and again, gets you most of the way there.

HOW LONG AN EXPOSURE EACH OPTION GIVES YOU

Pixel astro mode

tripod detected, automatic

~4 min

Manual camera app

full manual, RAW, iOS or Android

60s

Samsung Expert RAW

Pro mode, manual ISO

30s

iPhone Night mode

on a tripod, no manual control

30s

Any phone, handheld

the phone cannot hold still enough

3s

Longer is not always better. Four minutes gives a lovely Milky Way, but the stars will have visibly moved across the frame. Twenty to thirty seconds, repeated over and over, is the sweet spot for most people.

ON THE NIGHT

Clean the lens

A fingerprint you would never notice by day turns every bright star into a smeared halo.

Keep it firing

Take frame after frame for an hour or two. Every one is a Milky Way exposure, and the whole run is also a timelapse and a star trail.

Frame something solid

A rock, a dune ridge, a tree line, a friend looking up. Two minutes on composition pays for the whole session.

Screen down, red light on

Eyes take about twenty minutes to adapt and a bright screen undoes that in a second.

Power bank, and watch for dew

Long exposures drain a phone fast. If everything suddenly looks soft, wipe the lens.

TIMELAPSES AND STAR TRAILS

It is the same hour, played back

A star trail is that hour drawn as one picture. A timelapse is the same hour as video. Hyperlapse on a Samsung, Time-lapse on an iPhone, both happier once the phone is already on a tripod.

The exchange rate is brutal

Video runs at about 25 frames a second, so a satisfying ten second clip needs a few hundred separate exposures. That is an hour or two of real time for a few seconds of video.

Do not touch it once it is running

Picking the phone up to check ends the sequence, and there is no way to join the two halves back together afterwards. Set it going and walk away.

The full guide, with examples

Full walkthrough with example images, where the astro mode hides on each make, and the camera version of this card for when you want to go further than the phone can take you. There is a separate card for meteor showers.

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